



BSc (Hons) in Information Technology Interactive Media

Year 3 semester 1

IT3061 – User Experience Designing

Assignment 01

Student ID	Student Name	Mobile Number	Email
IT23642546	Bandara P.K.S	0714908688	it23642546@my.sliit.lk
IT23625266	Ranasingha R.A.T.C	0756941259	it23625266@my.sliit.lk
IT23675766	Loshana.A	0766875802	it23675766@my.sliit.lk
IT23827844	Fonseka W. A .T .D	0778147508	it23827844@my.sliit.lk
IT23829374	Abayanayaka A.A	0704978769	it23829374@my.sliit.lk

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1. Product Selection & UX Problem Definition

1.1. Selected Product

The selected home appliance for this project is a refrigerator. Refrigerators are essential household and commercial appliances used to store and preserve food by maintaining a low temperature. However, frequent opening of the refrigerator door causes cold air to escape, reducing energy efficiency and increasing electricity consumption.

1.2. Primary User Groups & Usage Context

1. Low Electricity-Bill Household Users

- These are the families that need a fridge but are not capable of affording to use the fridge as wish.
- They turn on the fridge only at nonpeak hours and try to preserve the cool throughout the day.

2. High Electricity Bill Household Users

- These are households that use refrigerators continuously due to daily needs such as food preservation and storage of temperature sensitive items like medicines.
- They experience high electricity bills mainly because the refrigerator runs constantly and frequently loses cold air when the door is opened.
- These users are unable to switch off the refrigerator and therefore require energy efficient solutions that reduce power consumption without affecting cooling performance.

3. Boarding House Students

- These are university students living in shared boarding houses with limited financial resources, making them highly sensitive to electricity costs.
- The refrigerator is shared among multiple users and opened frequently throughout the day, often leading to unnecessary cold-air loss.
- They require energy-efficient solutions that minimize power usage, prevent cooling loss, and support responsible shared usage without affecting food freshness.

4. Small Shop Owners

- Small grocery shops use commercial refrigerators continuously to keep beverages and perishable items fresh for customers.
- The refrigerator operates **24/7**, leading to high electricity usage.
- Frequent customer door openings cause significant cold-air loss.
- This cold-air loss forces the compressor to work harder, increasing electricity bills.
- Shop owners cannot turn off the refrigerator because cooling is essential for product safety and sales.
- They need energy-efficient solutions that reduce power consumption without affecting cooling performance or customer convenience.

5. Hotel / Restaurant Owner

- Restaurant owners use refrigerators continuously to store vegetables, meat, seafood, dairy products, desserts, and beverages required for daily operations.
- The refrigerator is opened very frequently by chefs and kitchen staff during busy hours, especially at lunch and dinner times. This causes cold air to escape repeatedly, reducing cooling efficiency.
- Because of this, the compressor has to work harder and longer to maintain the required temperature, leading to high electricity consumption and increased monthly utility costs.
- Restaurant owners cannot switch off or reduce the refrigerator usage because maintaining proper food temperature is essential for food quality, customer safety, and health regulations.
- In addition, unstable temperatures can spoil ingredients, create food waste, and cause financial losses.

1.3. UX Problems Identified

- Cold air loss when the refrigerator door is opened, causing temperature fluctuation.
- Energy inefficiency due to repeated cooling after cold air escapes.
- Users often keep the door open for a long time while searching for items.
- Lack of compartment-level access, exposing the entire fridge space unnecessarily.
- Poor feedback or indication about energy loss or door open duration.

1.4. UX & Usability Goals

- Effectiveness
 - Allow users to access stored items while minimizing cold-air loss.
- Efficiency
 - Reduce the cold-air loss as much as possible.
- Learnability
 - Ensure the new system is easy to understand and operate without training.
- Safety
 - Maintain stable internal temperatures to ensure food safety.
 - Implement safe to use interface and hardware components.
- Accessibility
 - Make the solution easy to use for different users including families, students, and workers.

2. UX-Focused Competitor Analysis

- Selected Appliance: Refrigerator
- Primary User Group: Household users
- Context of Use: Kitchen

UX Dimension	Product A – LG InstaView	Product B – Samsung Family Hub	Product C – Whirlpool Double Door	UX Insights
Physical Affordances	Glass knock panel, door compartments, LED controls	Large touchscreen display, handles, shelves	Physical temperature dial, door shelves	Physical controls are easier for quick tasks; touchscreens add complexity
Control - Action Mapping	Knock - view inside, open small door for quick access	Touchscreen - settings, apps, temperature	Turn dial - adjust cooling	Direct mechanical controls are easier to understand
Feedback & Visibility	Interior LED lighting, door open alarm	Screen notifications, visual indicators	Basic light and temperature indicator	Smart fridges provide richer feedback but can increase complexity
Learnability (First-time Use)	Easy - knock feature is intuitive	Medium - requires learning touchscreen interface	Very easy - traditional design	Simpler controls reduce learning time
Efficiency	Quick access door reduces full door opening	Many steps through touchscreen	Direct access but full door must open	Compartment access improves efficiency
Error Prevention & Recovery	Door alarm, temperature alerts	Smart alerts via mobile app	Basic door seal and alarms	Smart alerts improve error prevention
Accessibility	Large handles, visible shelves	Screen may be difficult for elderly users	Simple controls accessible for most users	Physical controls improve accessibility
Cognitive Load	Low simple interactions	High – many features and apps	Very low minimal interaction	Too many smart features increase cognitive load

Consistency Across Touchpoints	Consistent appliance + mobile app control	Ecosystem with smart home integration	Mostly physical interaction only	Smart ecosystems improve consistency
Packaging UX	Setup instructions and protective packaging	App setup and guided installation	Simple installation guide	Clear onboarding improves first-time experience
Marketing Video UX	Shows energy saving and quick access	Promotes smart lifestyle features	Focuses on reliability and storage	Marketing strongly shapes user expectations

3. User Research & UX Analysis

1st all survey was conducted to get basic ideas about the issues regarding using a fridge and electricity bill management. In this survey, random users were asked about their experience using a fridge and their pain points. Based on the result, we identified some user categories to interviewed to gather data for the research.

[Survey Link](#)

3.1. Low Electricity-Bill Household Users

3.1.1. Primary user research – Interview

To gather Data from different types of users, a user who has mid monthly electricity bill, was interviewed.

[Interview Link](#) (Click This)

3.1.2 Persona

Based on the data collected by user that has low electricity bill, below persona was created.



Nirmala Perera

SCHOOL TEACHER

OVERVIEW

School teacher living with her family in a middle-lower income household.

ABOUT

- Age - 53 Years
- Occupation - School Teacher
- Monthly Income - 160,000 Rs.
- Electricity Bill - 7,500 Rs.
- District- Gampaha
- Appliance Use - Standard home refrigerator

GOALS

- Minimize electricity consumption
- Preserve cold temperature when the fridge is switched off
- Reduce cold air loss when opening the fridge
- Keep food fresh for longer without increasing energy usage

MOTIVATION QUOTE

"We carefully manage electricity usage at home, so I want my fridge to help preserve the cold instead of losing it every time I open the door."

BEHAVIORS

- Uses the fridge only when necessary
- Opens the fridge quickly and carefully
- Plans fridge usage during non-peak hours
- Stores food in an organized manner to avoid long door openings

PAIN POINTS

- Cold air escapes quickly when the fridge is opened
- Stored cool temperature is lost easily during usage
- Fridge design does not support energy-saving habits
- Even short openings reduce cooling efficiency

NEEDS

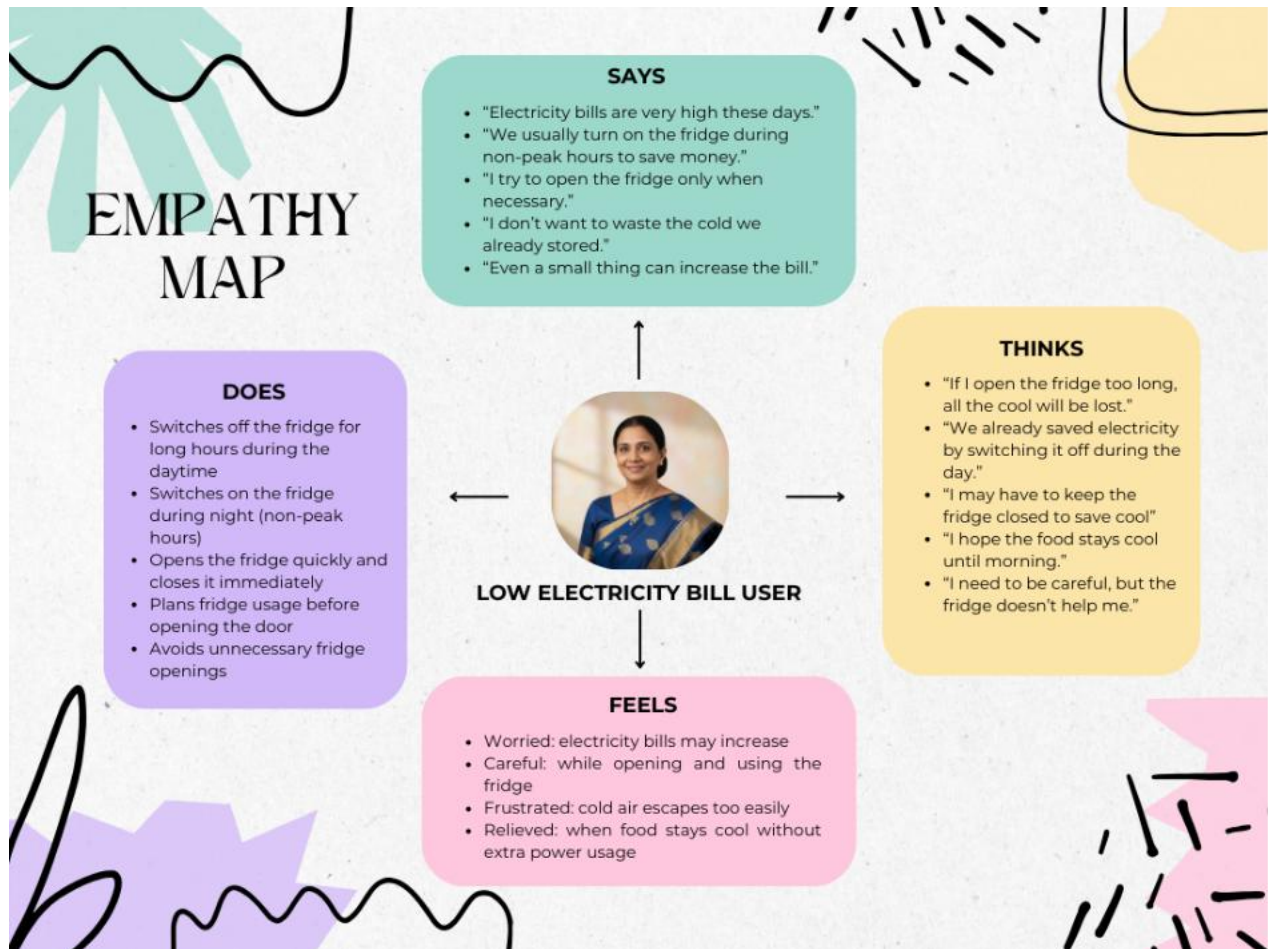
- A fridge design that preserves internal cool air
- A solution that works without increasing electricity use
- Minimal effort and no complex controls
- A reliable system that supports existing energy-saving behavior

+123-456-7890

hello@reallygreatsite.com

reallygreatsite.com

3.1.3. Empathy Map



3.1.4. User Journey Map



3.1.5. Key Findings

- Users are highly concerned about electricity consumption, especially in low-income households
- Many users switch off the fridge during the day to reduce electricity bills
- Users try to open the fridge quickly, but still experience cold-air loss
- The fridge is often kept open while searching for items, increasing energy loss
- Current refrigerator design does not support energy-saving behavior
- Users want convenience + energy efficiency without extra effort
- There is no internal mechanism to preserve cooling when the door is opened

3.1.6. Problem Statements

- Energy-conscious users need a way to preserve cold air when opening the fridge, because current designs cause energy loss and higher electricity usage
- Frequent fridge users need a way to access items quickly without exposing the entire compartment, because long door openings reduce cooling efficiency
- Users need a refrigerator that supports their behavior (quick access, low energy use), because current systems rely entirely on user discipline

3.1.7. HMW Questions

- How might we reduce cold-air loss when the fridge door is opened?
- How might we allow users to access only a part of the fridge instead of the whole space?
- How might we support energy-saving behavior without adding extra effort for the user?
- How might we help users find items quickly to reduce door-open time?
- How might we maintain cooling efficiency even with frequent usage?

3.1.8. User Stories

- As a home user, I want the fridge to preserve cold air when I open the door so that the stored cool is not wasted and electricity usage stays low.
- As a home user I want the fridge to retain cool temperature during usage after being switched off so that food stays fresh without turning the fridge on again.
- As a busy household user, I want the fridge to work without complicated interfaces so that I don't need to spend extra time for extra functionalities.
- As a home user I want the fridge to be opened for a while so that I can go through inventory and decide what to take.
- As a household user I want the fridge to always allow manual access so that I can still use it even if the system fails or power is off.

3.2. High Electricity Bill Household User Analysis

- High power consumption of residential users is the main subject of this study. Because of their way of life and domestic requirements, these people use refrigerators constantly. These consumers are worried about needless energy waste even when they are able to pay increased electricity costs. They favor energy saving options that have nothing to do with their comfort or regular use.

3.2.1. Primary user research – Interview

<https://drive.google.com/drive/folders/1UdNMBRKNoEg9o32UXuLuCAvcHN-FnJN5?usp=sharing>

3.2.2 Persona

To represent the intended user group, a persona was created using the information gathered. This persona represents a user who is financially secure but driven to cut down on electricity waste while maintaining stable and safe cooling for



OVERVIEW

CEO of an IT company living in Homagama with her family in a High income household.

ABOUT

- Age – 32 Years
- Occupation – CEO (IT Company)
- Family Status – Unmarried (Living with family)
- House Type – 3-bedroom house
- Monthly Income – Rs.700,000
- Electricity Bill – Rs.20,000
- Appliance Usage – Multiple IT devices and standard home appliances
- Refrigerator Usage – Runs continuously (24/7)

GOALS

- Reduce refrigerator electricity consumption
- Lower overall household electricity bill
- Prevent cold air loss when opening the fridge door
- Maintain stable cooling for food and medicines
- Improve internal insulation efficiency
- Keep medicines safe without increasing energy use

Randi Nimasha

HIGH ELECTRICITY BILL
HOUSEHOLD USER

MOTIVATION QUOTE

"I need a fridge that saves electricity without affecting medicine safety."

BEHAVIORS

- Uses desktop computers and laptops daily for work
- Keeps networking devices powered continuously
- Uses household appliances regularly
- Frequently opens refrigerator throughout the day
- Stores temperature-sensitive medicines inside the fridge
- Monitors electricity bill monthly

PAIN POINTS

- High monthly electricity bills
- Refrigerator runs continuously
- Cold air escapes when the door is opened
- Fear of medicine spoilage due to temperature fluctuation
- Current fridge lacks effective internal insulation

NEEDS

- A fridge design that preserves internal cool air
- A system that prevents cold air loss during door opening
- Stable and reliable cooling for medicines
- A simple energy-saving mechanism
- Reduced compressor workload to save electricity

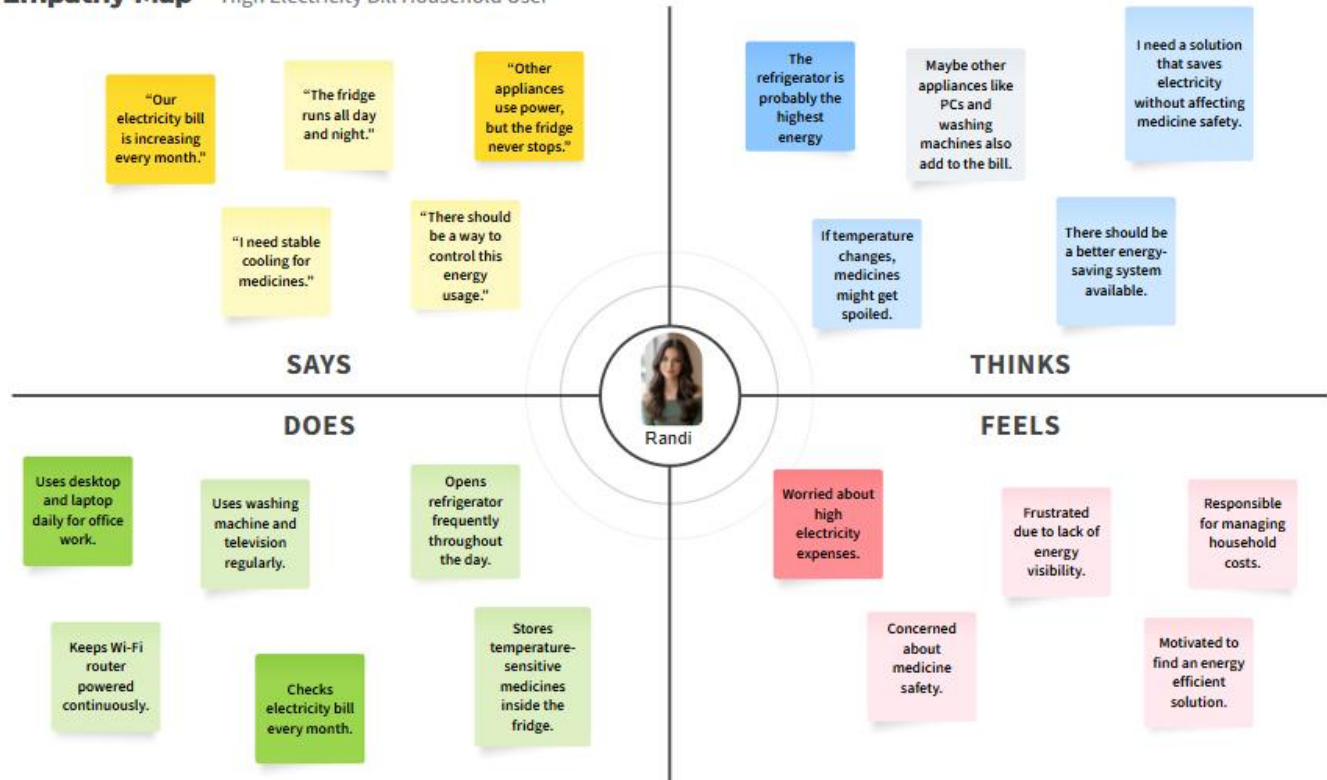
+94 750801356

contact@randinimasha.com

<https://randi.com>

3.2.3. Empathy Map

Empathy Map High Electricity Bill Household User



3.2.4. User Journey Map

USER JOURNEY MAP :- High Electricity Bill Household User

 USER INFO Name: Ms. Randi Nimasha Quote: "The fridge runs 24/7. I need to reduce electricity cost while keeping medicines safe."		SCENARIO Ms Randi notices her electricity bill is increasing. The refrigerator runs 24/7 to store food and medicines. She wants to reduce energy use without affecting cooling stability.		EXPECTATIONS 1.Reduce electricity bill 2.Maintain stable cooling 3.Prevent cold air loss	
STAGES	AWARENESS ▶	PROBLEM IDENTIFICATION ▶	SEARCHING FOR SOLUTION ▶	IMPLEMENTATION ▶	OUTCOME
GOALS	Understand why electricity bill is increasing	Identify main electricity consuming appliance.	Find an energy efficient solution.	Reduce refrigerator power consumption	Confirm electricity bill reduction.
ACTIONS	1. Receives monthly electricity bill 2. Reviews total amount 3. Discusses usage with family	1. Notices fridge cold air escaping when door opens 2. Compares usage with PCs and other appliances 3. Notices fridge never turns off	1. Searches for energy saving refrigerator designs 2. Looks for stable cooling features 3. Evaluates insulation improvements	1. Uses redesigned fridge 2. Inner polythene panel closes when door opens 3. Cold air remains inside	1. Receives next electricity bill 2. Compares previous and current bills
THOUGHTS	1. Why is the bill higher this month? 2. Which appliance is consuming the most power?	1. The fridge runs 24/7 2. This might be the highest energy consumer. 3. Cold air is escaping	1. I need energy saving without medicine risk 2. Cooling must remain stable	1. Cool air is not escaping now 2. Energy loss is reduced	1. Electricity cost has reduced 2. Medicines are safely stored
PAIN POINTS	1. No clear breakdown of appliance wise energy use 2. Confusion about high consumption	1. Continuous refrigerator operation 2. No real time monitoring system	1. Fear of temperature fluctuation 2. Uncertainty about real energy savings	1. Need simple mechanism 2. Must ensure durability	1. Maintain long term performance
EMOTIONS	 Interested	 Annoyed	 Frustrated	 Happy	 Happy
TOUCHPOINTS	1. Electricity bill 2. Family discussion	1. Refrigerator 2. Home appliances	1. Online research 2. Product comparisons	1. Refrigerator door 2. Inner cooling barrier system	1. Electricity bill 2. Refrigerator system
OPPORTUNITIES	1. Identify refrigerator as main energy source 2. Highlight cold air loss impact	1. Real time energy monitoring display 2. Reduce cold air loss physically	1. Introduce internal polythene cooling barrier 2. Add automatic inner panel system 3. Improve insulation design	1. Enhance insulation efficiency 2. Combine with energy monitoring display	1. Promote sustainable energy saving 2. Improve long term efficiency

3.2.5. Key Findings

The following results have been made from the examination of user data:

- Refrigerators are among the appliances that use the most energy as they run constantly.
- When the refrigerator door is opened, a large amount of cold air escapes.
- Despite being aware of energy waste, users are unable to cut back because of necessary necessities.
- Even consumers with expensive bills would rather cut back on needless electricity use.
- For the storage of medications that are sensitive to temperature, stable cooling is essential.
- There are currently no systems in place to stop cold air loss while using refrigerators.

3.2.6. Problem Statements

- Due to the refrigerator's constant operation and lack of effective energy-saving features, users find it difficult to lower its energy usage.
- Because cold air escapes when the refrigerator door is opened, users waste energy.
- Customers require an energy-saving solution that doesn't compromise cooling efficiency or usability.

3.2.7. HMW Questions

- How might we reduce unnecessary refrigerator energy consumption while maintaining continuous operation?
- How might we prevent cold air loss when the refrigerator door is opened?
- How might we maintain stable cooling for food and temperature-sensitive medicines?
- How might we design a simple and practical energy saving solution for everyday user?

3.2.8. User Stories

- As a high electricity bill household user, I want to reduce refrigerator energy consumption so that my monthly electricity expenses decrease.
- As a user storing temperature sensitive medicines, I want stable internal cooling so that medicines and food remain safe.
- As a user, I want to prevent cold air from escaping when the fridge door opens so that energy is not wasted.
- As a user, I want an internal polythene cooling barrier so that cold air stays inside even when accessing items.
- As a user, I want the inner cooling panel to close automatically when the door opens so that cooling efficiency is maintained.
- As a user, I want improved internal insulation so that the compressor works less and saves electricity
- As a user, I want simple and easy-to-use controls so that I can operate the refrigerator without confusion.

3.3 Small Grocery Shop Owner Analysis

Small grocery shop owners rely on commercial refrigerators that run continuously to keep products fresh. Frequent door openings by customers cause cold-air loss, high electricity consumption, and unstable cooling.

These users want to reduce unnecessary energy waste while still maintaining strong cooling performance for business needs.

3.3.1 Primary User Research – Interview

[0330.mov](#)

3.3.2 Persona

To represent the intended user group, a persona was created using the information gathered. This persona represents a user who is financially secure but driven to cut down on electricity waste while maintaining stable and safe cooling for



OVERVIEW

Sunil ariyapala runs a small grocery shop where he uses a commercial refrigerator 24/7 to keep beverages, dairy, and perishables fresh for his customers.

ABOUT

- Age -41 Years
- Occupation – Grocery Shop Owner
- Shop Type – Small local retail store
- Electricity Bill – High due to commercial fridge usage
- Products Stored – Soft drinks, milk, yogurt, chocolates.
- Refrigerator Usage – Runs continuously (24/7)

GOALS

- Keep products fresh and safe for customers
- Reduce electricity costs for his shop
- Prevent cold air loss during frequent door openings
- Maintain consistent cooling temperature
- Avoid product spoilage

sunil ariyapala

GROCERY SHOP
REFRIGERATOR USER

MOTIVATION QUOTE

“My fridge must keep products cold all day, but I don’t want it increasing my electricity cost.”

BEHAVIORS

- Opens refrigerator frequently throughout the day
- Restocks products regularly
- Keeps fridge running continuously
- Monitors electricity bill monthly
- Adjusts temperature during peak hours

PAIN POINTS

- High electricity bills due to continuous operation
- Cold air escapes because customers open the fridge often
- Compressor works harder during busy hours
- Fear of product spoilage during power cuts
- Uneven cooling when door is opened frequently

NEEDS

- A refrigerator that retains cold air during frequent opening
- Strong internal insulation
- Stable temperature control
- Energy-efficient compressor system
- Durable design for heavy daily use

+94 78675644

contact@ariyapala.com

<https://ari.com>

3.3.3. Empathy Map

Empathy Map of Grocery Shop Refrigerator User



3.3.4. User Journey Map

User Journey Map



USER INFO

Name: Mr. Sunil ariyapala

Quote: "The fridge must keep products cold all day without increasing my electricity bill."

SCENARIO Mr. Sunil operates a small grocery shop where customers frequently open the refrigerator. The fridge runs continuously, causing high electricity consumption and energy loss due to repeated door openings. EXPECTATIONS • Maintain stable cooling • Prevent cold air loss • Reduce electricity cost • Avoid product spoilage		STAGE 1	STAGE 2	STAGE 3	STAGE 4	STAGE 5
	GOAL	Recognize high electricity consumption	Identify cause of energy loss	Find energy-saving refrigerator design	Reduce electricity usage	Confirm cost reduction
	ACTIONS	• Receives monthly electricity bill • Notices high charges	• Observes customers opening fridge often • Notices cold air escaping	• Looks for improved commercial fridge models • Evaluates insulation options	• Uses redesigned fridge • Cooling barrier reduces cold air loss	• Reviews next electricity bill • Compares profit margin
	THOUGHTS	"The fridge is using too much power."	"Frequent openings increase energy loss."	"I need strong cooling with less power usage."	"Energy loss has reduced."	"Electricity cost is lower."
	PAIN POINTS	No energy efficiency in current system	Continuous compressor operation	Fear of product spoilage	Installation cost is high	Need regular maintenance
	EMOTIONS	Concerned	Annoyed	Frustrated	Satisfied	Happy
	OPPORTUNITIES	Improve insulation system	Add internal cooling barrier	Introduce automatic inner panel system	Enhance insulation efficiency	Promote sustainable energy-saving system

3.3.5 Key Findings

- Commercial fridges consume high energy due to continuous usage.
- Cold-air loss is the main cause of extra electricity consumption.
- Shop owners cannot compromise cooling because products must stay fresh.
- Frequent door openings increase compressor workload.
- Cold-air protection systems are not included in standard fridges.

3.3.6 Problem Statements

- Shop owners cannot reduce refrigerator power usage because it must run 24/7.
- Cold air escapes during frequent customer door openings, causing energy loss.
- No mechanism exists to maintain cooling efficiency during busy hours.
- Shop owners need an energy-saving method without affecting usability or cooling performance.

3.3.7 HMW Questions

- How might we reduce refrigerator energy usage without stopping continuous operation?
- How might we prevent cold-air loss when the fridge door is opened by customers?
- How might we maintain stable cooling during peak hours?
- How might we design an affordable, practical energy-saving solution?

3.3.8 User Stories

- As a small grocery shop owner, I want to reduce refrigerator energy consumption so that my business profit increases.
- As a shop owner, I want stable cooling so that perishable items remain fresh.
- As a user, I want to prevent cold air from escaping during customer use.
- As a user, I want an internal cooling barrier so that cold air stays inside.
- As a user, I want the panel to close automatically for efficiency.
- As a user, I want better insulation so the compressor works less.
- As a user, I want simple controls to manage the refrigerator easily.

3.4. Boarding House Student Analysis

Boarding house students with limited financial resources are a key focus of this study, as they are highly affected by electricity costs. Due to shared living conditions and daily usage needs, refrigerators are used frequently by multiple users.

These users are concerned about unnecessary energy waste, especially when the refrigerator door is left open or misused by others. They prefer energy-efficient solutions that help reduce electricity consumption without affecting food freshness or their daily convenience.

3.4.1. Primary user research – Interview

<https://drive.google.com/file/d/1zeUF09LKQWDLE2-Q7klvmt8iXj3RpOng/view?usp=sharing>

3.4.2. Persona

To represent the intended user group, a persona was created using the information gathered. This persona represents a user who has limited financial resources and is highly concerned about reducing electricity costs while maintaining proper cooling for daily food storage in a shared living environment.



Sasindu Fernando

BOARDING STUDENT

OVERVIEW

A university student living in a shared boarding house with limited financial resources. Budget conscious and concerned about reducing electricity costs while maintaining food freshness.

ABOUT

- Age -23
- Occupation -Undergraduate Student
- Living Situation -Lives in a shared boarding house with other students
- Income Level: Low / Limited monthly budget
- Electricity Bill: Rs.5000
- Product Used: Single door refrigerator

GOALS

- Reduce monthly electricity expenses.
- Keep food and beverages fresh.
- Share the refrigerator without wasting energy.
- Prevent unnecessary cooling loss.

MOTIVATION QUOTE

"Reducing electricity usage helps manage monthly expenses effectively."

BEHAVIORS

- Opens the refrigerator multiple times a day.
- Sometimes keeps the door open while deciding what to take.
- Reminds roommates to close the refrigerator door.
- Checks electricity bills monthly.
- Prefers appliances with energy-saving features.

PAIN POINTS

- High electricity bills in the boarding house.
- Cool air escaping when the door remains open.
- Food spoiling faster due to temperature fluctuations.
- Lack of energy monitoring features.

NEEDS

- A feature that prevents cool air from escaping when the door is open.
- Energy efficient refrigerator operation.
- Affordable and cost saving appliance.
- Clear energy usage feedback.

+94 772357341

sasindufernando@gmail.com

<https://www.webpagetest.org>

3.4.3. Empathy Map

Empathy Map Boarding Student

Says

- The electricity bill is too high this month.
- Please close the fridge door properly.
- Food gets spoiled quickly.

Thinks

- I need to save money every month.
- The refrigerator uses too much electricity
- Cooling should not be wasted when the door is open.



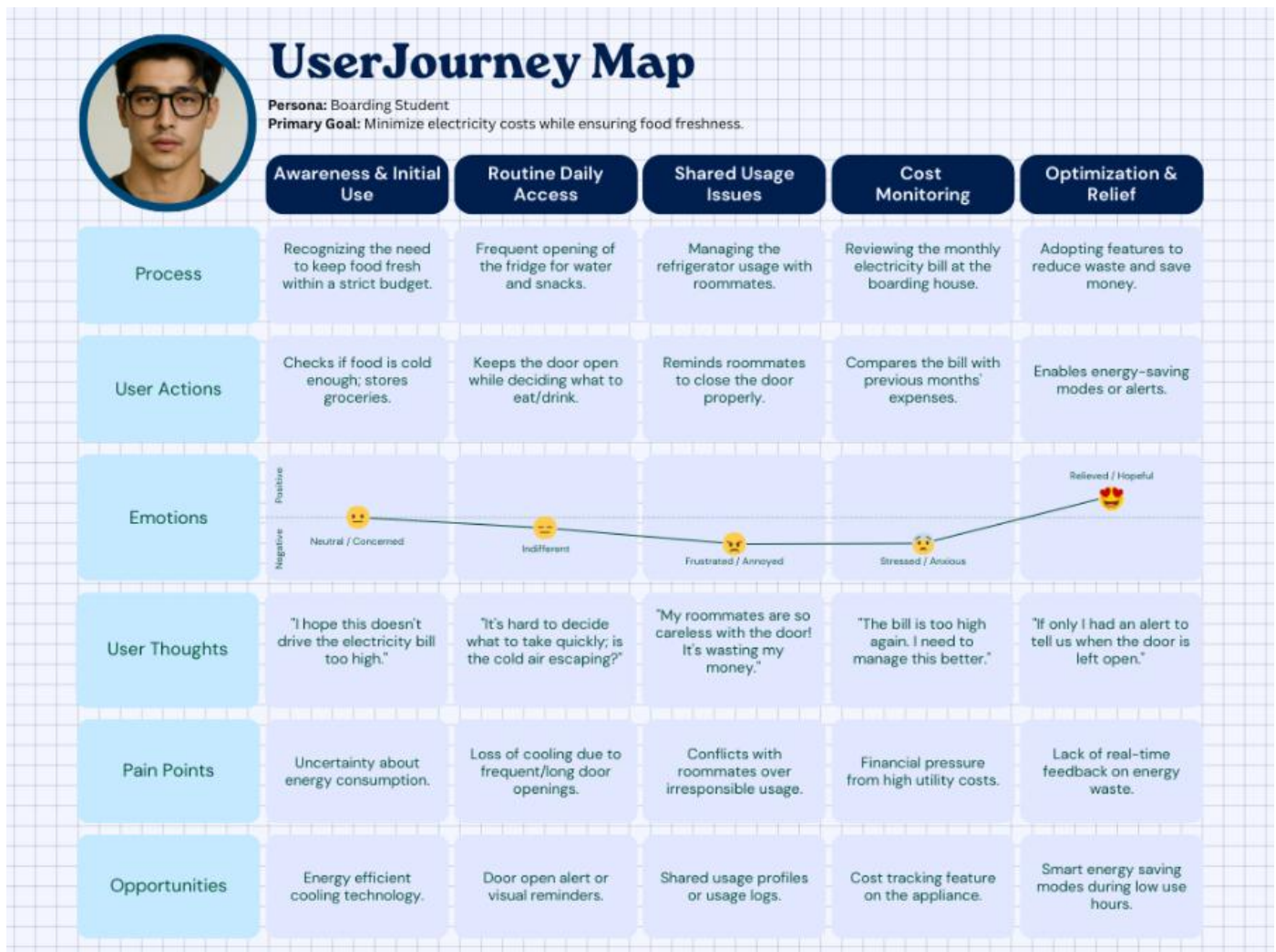
Does

- Shares the refrigerator with other students
- Opens the refrigerator many times a day.
- Tries to reduce electricity usage.
- Checks monthly electricity bills.

Feels

- Stressed when electricity bills are high.
- Frustrated when others misuse the refrigerator.
- Relieved when costs are reduced.

3.4.4. User Journey Map



3.4.5. Key Findings

The following results have been made from the examination of user data:

- Refrigerators are used frequently throughout the day due to shared usage among multiple boarding students.
- When the refrigerator door is kept open for longer periods, a significant amount of cold air escapes.
- Shared usage often leads to careless behavior, increasing unnecessary energy consumption.
- Boarding students are highly sensitive to electricity costs due to limited financial resources.
- There are no proper systems to monitor or prevent energy waste in shared refrigerator usage.

3.4.6. Problem Statements

- Due to frequent and shared usage of the refrigerator, boarding students find it difficult to control and reduce electricity consumption.
- Energy is wasted when the refrigerator door is left open or used inefficiently by multiple users.
- Boarding students require an affordable energy-saving solution that maintains cooling efficiency while supporting shared usage.

3.4.7. HMW Questions

- How might we reduce refrigerator energy consumption in a shared boarding environment?
- How might we prevent cold air loss when the refrigerator door is frequently opened by multiple users?
- How might we maintain stable cooling to prevent food spoilage in shared usage conditions?
- How might we design a simple and cost-effective energy-saving solution suitable for boarding students?

3.4.8. User Stories

- As a boarding student, I want the refrigerator to prevent cool air from escaping when the door is open, so that electricity consumption is reduced.
- As a boarding student, I want an energy-saving refrigerator, so that I can manage my monthly expenses better.
- As a boarding student, I want an alert when the refrigerator door is left open for too long, so that energy is not wasted.
- As a boarding student, I want the refrigerator to support shared usage efficiently, so that all roommates use it responsibly.
- As a boarding student, I want the refrigerator to have compartments and shelves, so that I can organize my food efficiently and avoid spoilage.

3.5. Restaurant Owner Analysis

Restaurant owners use refrigerators all day to store food and drinks. The fridge is opened very frequently by kitchen staff, especially during busy hours.

This causes cold air to escape, making the fridge work harder and increasing electricity bills.

They cannot turn off the refrigerator because it is essential for food safety and daily operations.

3.5.1. Primary user research – Interview

https://drive.google.com/file/d/1nss74jomBlcIMrUpN8to3KEt8e6V0fYv/view?usp=drive_link

3.5.2. Persona

To represent the intended user group, a persona was created using the information gathered. This persona represents a restaurant owner who relies heavily on refrigeration for daily operations and is focused on reducing electricity costs while ensuring consistent cooling and food safety despite frequent usage.



Nimal Perera

RESTAURANT OWNER

OVERVIEW

A restaurant owner who depends heavily on refrigeration to maintain food quality while controlling operational costs, especially electricity expenses.

ABOUT

- Name: Nimal Perera
- Age: 40
- Occupation: Restaurant Owner
- Location: Colombo
- Business Type: Medium scale restaurant
- Income Level: Upper-middle income (stable monthly business revenue with high operational expenses)
- Electricity Bill: Rs 35000 - Rs 40000 (depending on season and business volume)
- Product Used: Double Door Refrigerator

GOALS

- Keep food fresh and safe for customers.
- Reduce monthly electricity bills.
- Minimize food spoilage and waste.
- Maintain smooth kitchen workflow.
- Ensure equipment durability and efficiency.

MOTIVATION QUOTE

"I need reliable equipment that keeps my ingredients fresh without increasing my electricity costs."

BEHAVIORS

- Refrigerators are opened very frequently during peak hours.
- Stores large quantities of meat, vegetables, and dairy products.
- Regularly monitors electricity bills.
- Invests in appliances that improve efficiency.
- Trains staff to handle equipment properly.

PAIN POINTS

- High electricity bills due to continuous refrigerator usage.
- Cool air escaping when the door is opened frequently.
- Temperature fluctuations affecting food freshness.
- Risk of food spoilage during busy hours.
- Staff sometimes leaving the door open longer than necessary.

NEEDS

- A refrigerator that prevents cool air from escaping when the door is open.
- Faster temperature recovery after closing the door.
- Energy efficient cooling system.
- Door open alert system.
- Durable appliance suitable for commercial use.

+94 781356221 nimalperera10@gmail.com <https://www.nimalrestaurant.lk>

3.5.3. Empathy Map

Empathy Map

Restaurant Owner

SAY

- Electricity costs are too high.
- We open the fridge constantly during rush hours.
- Food quality must never drop.

THINK

- If temperature changes, food might spoil.
- Energy waste is affecting my profit.
- I need more efficient equipment.

DOSE

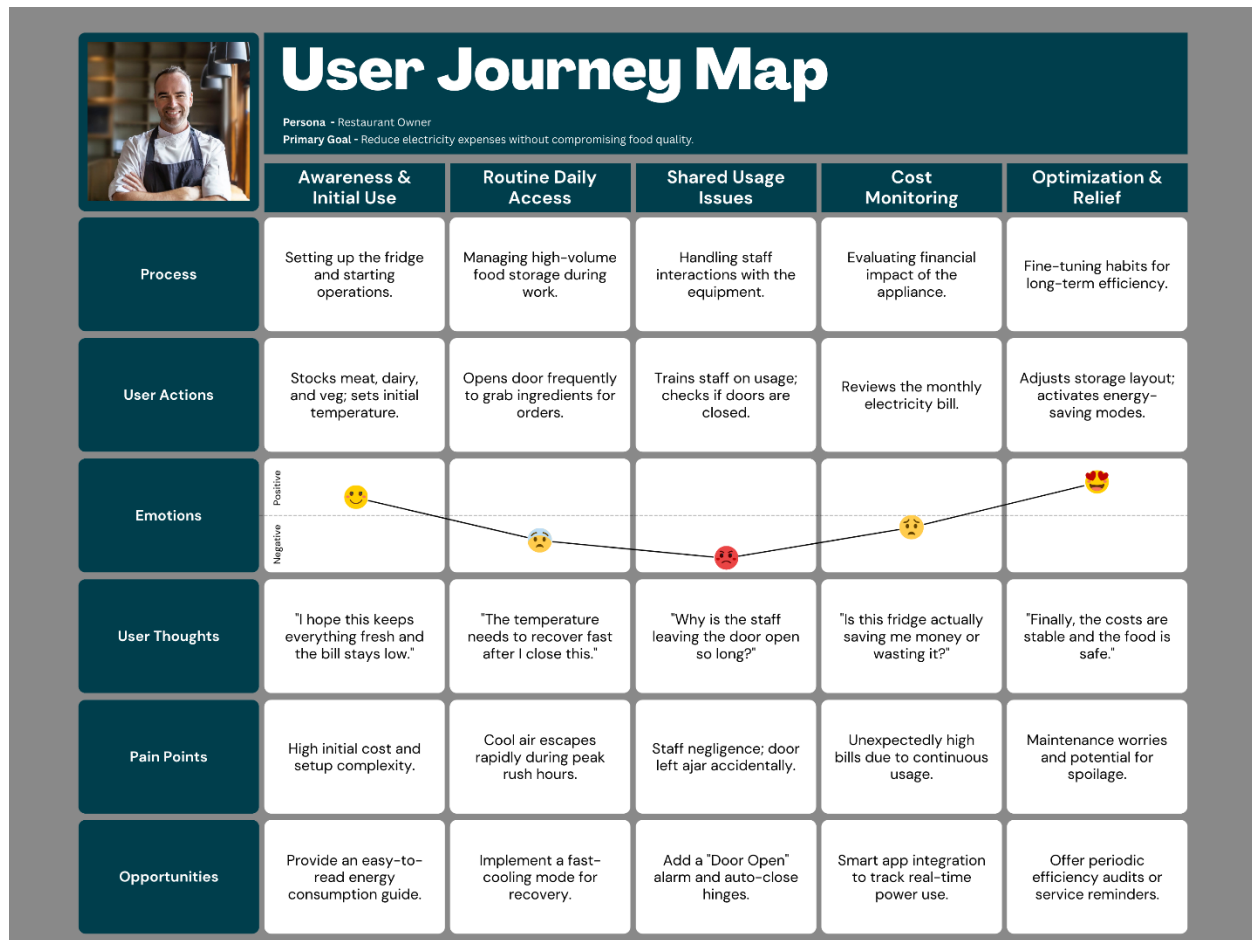
- Monitors electricity consumption.
- Checks food storage conditions daily.
- Trains staff on proper fridge usage.
- Invests in reliable kitchen equipment.

FEELS

- Stressed about operational costs.
- Concerned about food safety.
- Frustrated when energy bills increase.
- Responsible for maintaining quality.



3.5.4. User Journey Map



3.5.5. Key Findings

The following results have been made from the examination of user data:

- Refrigerators are used continuously throughout the day due to high kitchen demand and food storage needs.
- Frequent opening of the refrigerator door during busy hours leads to significant cold-air loss.
- Kitchen staff sometimes leave the door open longer than necessary, increasing energy consumption.
- Restaurant owners are highly concerned about rising electricity costs due to continuous appliance usage.
- Temperature fluctuations can negatively impact food freshness and lead to spoilage.
- There are limited systems in place to monitor door usage or prevent cooling loss in commercial kitchens.

3.5.6. Problem Statements

- Restaurant owners struggle with high electricity consumption because refrigerators are used continuously and frequently opened during busy hours.
- Frequent door openings cause cold air to escape, making it difficult to maintain a stable internal temperature.
- Kitchen staff behavior, such as leaving the refrigerator door open longer than necessary, increases energy waste.
- Temperature fluctuations inside the refrigerator can lead to food spoilage and financial losses.
- There is a lack of smart systems to monitor usage and prevent unnecessary cooling loss in commercial environments.

3.5.7. HMW Questions

- How might we reduce cold-air loss when the refrigerator door is frequently opened?
- How might we help restaurant staff use the refrigerator more efficiently?
- How might we maintain a stable temperature inside the refrigerator during busy hours?
- How might we reduce electricity consumption without affecting cooling performance?
- How might we alert users when the refrigerator door is left open too long?
- How might we design a refrigerator that supports high-frequency use in restaurants?

3.5.8. User Stories

- As a restaurant owner, I want the refrigerator to prevent cool air from escaping when the door is open, so that electricity consumption is reduced.
- As a restaurant owner, I want the refrigerator to quickly restore its internal temperature after closing the door, so that food remains fresh and safe.
- As a restaurant owner, I want an alert when the refrigerator door is left open, so that energy is not wasted.
- As a restaurant owner, I want an energy-efficient commercial refrigerator, so that I can reduce operational costs.
- As a restaurant owner, I want stable temperature control,so that food safety standards are maintained.

4. UX Design & Prototyping

4.1. Proposed Solution

The proposed refrigerator design introduces an internal compartment system with automatic covers and a control panel interface. This system allows users to access specific sections of the refrigerator without exposing the entire interior, thereby reducing cold air loss and improving energy efficiency

4.2. Key Design Features

- Internal automatic covers for each section
- Section based access system
- User friendly control panel
- Organized storage compartments

4.3. Interaction Logic

The system operates as follows:

1. When the refrigerator door is opened, all internal covers remain closed
2. The user selects a desired section using the control panel
3. The selected section opens automatically
4. The user retrieves the item
5. When the door is closed, all sections return to their default state.

4.4. Design Justification

The design decisions were made based on identified UX problems:

1. Cold air loss reduction - Only one section opens
2. Improved efficiency - Faster access to items
3. Reduced cognitive load - Organized sections
4. Better control mapping - Clear user interaction via control panel

4.5. UX Improvements

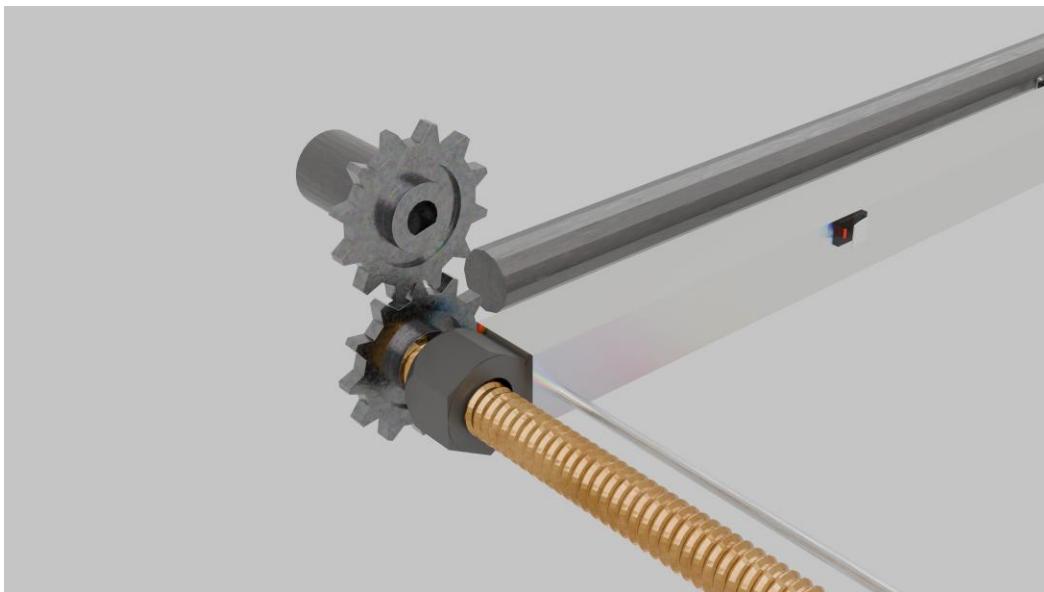
Compared to existing refrigerators, the new design:

1. Reduces energy consumption
2. Improves usability
3. Enhances user satisfaction
4. Provides better interaction feedback

4.6. 3D Prototype

Video Link: [Prototype Animation](#)





5. UX/Usability Evaluation & Iteration

5.1. Evaluation Method

The usability evaluation was conducted using a scenario-based approach, combining both a functional prototype and a 3D conceptual model.

Due to development limitations, only the control panel interaction (single-button prototype) was implemented as a working model, while the remaining system features were represented through a 3D prototype.

This hybrid approach allowed evaluation of both actual user interaction and proposed system behavior.

5.2. Evaluation Setup

Working Prototype:

A simplified control panel with a single functional button

3D Prototype:

Visual representation of the full refrigerator system including:

- Internal cover mechanism
- Section-based containers
- Full interaction design

5.3. Evaluation Scenario

A user interacts with the refrigerator to retrieve an item:

1. The user opens the refrigerator door
2. The user observes the control panel
3. The user presses the available button (prototype interaction)
4. The system demonstrates section-based access (via 3D model explanation)
5. The user understands how only one section opens
6. The user retrieves the item and closes the door

5.4. Evaluation Criteria

The evaluation focused on:

- Ease of use
- Efficiency
- Error prevention
- Energy efficiency
- User satisfaction

5.5. Evaluation Results

1. Ease of Use

Users were able to interact with the prototype button easily. The simple interaction helped users understand the concept quickly.

2. Efficiency

The concept of opening only one section (shown through 3D model) was perceived as faster and more efficient than traditional usage.

3. Error Prevention

Users understood that the system prevents unnecessary opening of sections, reducing mistakes.

4. Energy Efficiency

The internal cover concept (demonstrated through 3D visualization) was clearly understood as a method to reduce cold air loss.

5. User Satisfaction

Users responded positively to the idea of controlled access and energy-saving design.

5.6. Identified Limitations

- Only one button was physically implemented in the prototype
- Full interaction flow was not fully functional in real hardware
- Some features were explained using 3D visualization instead of real interaction

5.7. Suggested Improvements

- Develop a fully functional multi-button control panel
- Integrate real automatic internal cover mechanism
- Add visual indicators and feedback system
- Improve real-time interaction between sections

5.8. Evaluation Summary

The evaluation demonstrates that even with a partially functional prototype, users were able to understand and interact with the system concept effectively.

The combination of a working control element and a 3D model helped validate the usability and feasibility of the design.

The proposed system shows strong potential to improve efficiency, reduce energy loss, and enhance overall user experience.

6. UX Communication Artefacts

The following artefacts were developed to communicate the UX design and functionality of the proposed refrigerator system:

- User Manual – Provides step-by-step instructions on how to use the system
- 3D Design Visuals – Demonstrates the internal structure and interaction flow
- Control Panel Interface – Shows how users interact with different sections
- Marketing Materials – Illustrate the problem, solution, and usability improvements

These artefacts help users understand the system, improve usability, and support correct interaction with the product.

6.1. User Manual



More Details Link: [User Manual.pdf](#)

6.2. Flyer

[flyer front page.jpeg](#)

[flyer back page 2.jpeg](#)

NO MORE WASTED COOL AIR.

INTRODUCING THE ELORA 'ACTIVE SEAL' SERIES, ACCESS ANY COMPARTMENT, KEEP THE REST OF YOUR COOL LOCKED IN.

LOCK IN THE COLD. SHUT OUT THE COST.

Featuring our revolutionary 'Active Seal' Automatic Inner Compartment Barriers, the next level of ELORA technology for all user groups.





10 YEAR PREMIUM GUARANTEE



COMPARTMENT-LEVEL ACCESS.
Open only what you need. The rest stays sealed.



PRECISE TEMP FOR MEDS & FRESH FOOD.
Prevents temperature fluctuations.



EXTENDED RETENTION & MANUAL ACCESS.
Manual access always, on or off, and extended cool-up to 12 hours when off.



UP TO 35% LOWER ELECTRICITY COSTS.
Saves energy for families, shops, students, and restaurants.



AUTOMATIC ACTIVE SEAL BARRIER.
Deploys automatically when the outer door opens. Total defense against cold-air loss.



SCAN FOR INTERACTIVE EXPERIENCE

ELORA ACTIVE SEAL SERIES: ENGINEERING EXCELLENCE.

Hot line: 0766875802
0112173279

Elora Series Models: • Models: [ELORA-300, ELORA-250, ELORA-300]
• Capacity: Total - Fridge - Freezer
• Annual Energy Consumption: 180-220 kWh/year
• Noise Level: ~ 50-60 dBA
• Coolant Type: (Non-Toxic)

ELORA

THE SCIENCE BEHIND THE SEAL: TOTAL PRESERVATION.



1. USER REQUESTS COMPARTMENT

2. ACTIVE SEAL BARRIER DEPLOYS

3. SPECIFIC ACCESS WITHOUT COOL-AIR LOSS

4. SPECIFIC ACCESS WITHOUT COOL-AIR PROTECTION

4. TOTAL DEPRESSURIZATION PROTECTION



MEDICAL & RESEARCH READY
Vaccine vials & syringe and microscope.



COMMERCIAL CATERING EFFICIENCY
Saves energy for imlis, restaurants.



MAXIMIZED FRESHNESS FOR PRODUCE
Cornucopia vegetables.



BULK FOOD PRESERVATION
Frozen foods, stack foods, and warehouse.

FEATURE	UNIT	ELORA-300	ELORA-300	ELORA-250	ELORA-300
Dimensions (H x W x D)	mm	790 x 290 x 250	270 x 250 x 250	270 x 250 x 250	790 x 290 x 300
Individual Compartment Vol.	m³	2.20	2.50	2.30	2.70
Individual Compartment Vol.	m³	1.70	1.30	1.20	1.50
Individual Compartment Vol.	m³	0.50	0.30	0.50	0.30
Compressor Type	-	Inverter			
Warranty	-	Expanded: 2 Year warranty no ho late for 10 warranty)			
Noise Level (Speed)	Speed	25-30	25-40	25-40	30-40
Noise Level (Speed)	Speed	30-40	30-40	30-40	30-50
Noise Level (Speed)	Speed	40-60	40-60	40-60	40-60
Annual Energy Consumption:					
- User Profile	kWh/rh	180-200	80-120	80-120	180-220
- Unocal Energy Consumption	kWh/rh	0.30	0.20	0.80	1.20
- User Profile - User Profile	kWh/rh	0.30	0.80	0.90	1.20



10 YEAR PREMIUM GUARANTEE

TECHNICAL SPECIFICATIONS

• Models: [ELORA-300, ELORA-250, ELORA-300]
• Capacity: Total - Fridge - Freezer
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• Noise Level: ~ 50-60 dBA
• Coolant Type: (Non-Toxic)

Hot line: CONTACT US
0766875802
0112173279

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6.3. Brochure

[Brochure.jpeg](#)

ELORA

NO MORE WASTED COOL AIR. LOCK IN THE COLD. SHUT OUT THE COST.

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COMPARTMENT-LEVEL ACCESS.
Open only what you need. The rest stays sealed.

UP TO 35% LOWER ELECTRICITY COSTS.
Saves energy for families, shops, students, and restaurants.

PRECISE TEMP FOR MEDS & FRESH FOOD.
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Deploys automatically when the outer door opens. Total defense against cold-air loss.

EXTENDED RETENTION & MANUAL ACCESS.
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BULK FOOD PRESERVATION FCQNI.
Frozen foods, for imlis, restaurants.

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Vaccine vials & syringe and microscope.

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MAXIMIZED FRESHNESS FOR PRODUCE.
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ELORA SERIES

TECHNICAL SPECIFICATIONS & ENGINEERED EXCELLENCE

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Dimensions (H x X x A)	mm	790 x 290 x 270	270 x 250 x 250	270 x 250 x 250	270 x 290 x 300
Individual Compartment Mt.	m ²	2.50	2.50	2.30	2.70
Individual Compartment Mt.	m ¹¹	1.50	1.50	1.50	1.50
Individual Compartment Mt.	m ¹¹	0.50	0.50	0.50	0.50
Compressor Type	-	Inverter			
Compressor Type	-	Inverter			
Warranty	-	Expanded: 2 Non-tempest (on to tede for 10 months)			
Noise Level	-	1 (ftm taels)			
Noise Level (Speed)	Speed	25-30	25-40	25-40	30-90
Noise Level (Speed)	Speed	30-40	30-40	30-40	30-90
Noise Level (Speed)	Speed	40-60	40-80	40-80	40-60
Annual Energy Consumption	- User Profile	180-300	80-120	80-120	180-300
Annual Energy Consumption	- Annual Energy Consumption	1.80	0.80	0.80	1.20
Annual Energy Consumption	- User Profile - User Profile	0.30	0.30	0.90	1.20

10 YEAR PREMIUM GUARANTEE

DURABLE & RELIABLE.
Built to perform, built to last.

ENERGY SMART.
Lower bills, greener future.

EXPERT SUPPORT.
We're here, always.

ENGINEERED FOR EXCELLENCE.
Innovation in every detail.

TECHNICAL SPECIFICATIONS

- Models: [ELORA-300, ELORA-250; ELORA-3000, ELORA-300] • Capacity: Total - Frigde
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SMART ACCESS. MAXIMUM PRESERVATION. MINIMUM WASTE. MAXIMUM VALUE.

LOCK IN THE COLD. SHUT OUT THE COST. CHOOSE ELORA ACTIVE SEAL SERIES.

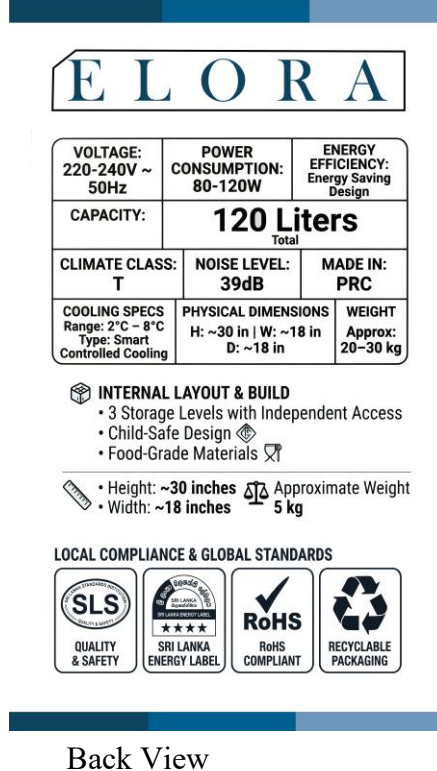
Hot line:
0766875802
0112173279

6.4. Marketing Video

[UED marketing video.mp4](#)

6.5. Package design

Link : [packaging design images](#)



7. Contribution Matrix

Student Declaration I confirm that the contribution percentage stated below accurately reflect my contribution to the assignment.

A. Contribution Matrix (Student-Filled)

1. IT23642546 - Bandara P.K.S
2. IT23625266 - Ranasingha R.A.T.C
3. IT23675766 - Loshana.A
4. IT23827844 - Fonseka W. A .T .D
5. IT23829374 - Abayanayaka A.A

Task/Deliverable	Student 1-ID	Student 2-ID	Student 3-ID	Student 4-ID	Student 5-ID
UX Problem Definition & Goals	20%	20%	20%	20%	20%
UX Competitor Analysis	20%	20%	20%	20%	20%
User Research (Data Collection)	20%	20%	20%	20%	20%
UX Analysis Artefacts (Personas, Journey, HMW, etc.)	20%	20%	20%	20%	20%
High-Fidelity Prototype (Physical + 3D)	0%	10%	30%	30%	30%
Packaging UX Design	50%	50%	0%	0%	0%
Printed Marketing Materials	0%	0%	100%	0%	0%
Marketing/Communication Video	50%	50%	0%	0%	0%
User Manual	10%	90%	0%	0%	0%
Usability & UX Evaluation	0%	10%	30%	30%	30%
Iteration & Refinement	0%	10%	30%	30%	30%
Report Writing & Compilation	0%	100%	0%	0%	0%
Presentation & Pitching	20%	20%	20%	20%	20%
Individual Contribution Percentage (C)					
Individual Mark					

Academic Integrity & Expectations AI tools used:

1. Chatgpt :
Get ideas and details about the documentation.
2. Google flow & Whisk
Creating Marketing video using these platforms.